

Yao-An Yang

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Education

- **University of Michigan** **Michigan, USA**
 - Bachelor of Science in Computer Science and Physics* *September 2022 – May 2026 (Expected)*
 - Cumulative GPA: 3.97/4.0 (CS major: 3.97; Physics major: 3.97)
 - Selected Coursework: Linear Algebra, Continuous Optimization Methods, Operating Systems, Data Mining, Natural Language Processing, Machine Learning, Computer Vision, Introduction to AI (in progress)

Publications

- Donald Loveland, **Yao-An Yang**, Danai Koutra. 2025. Glance for Context: Learning When to Leverage LLMs for Node-Aware GNN-LLM Fusion. Under Review at the International Conference on Learning Representations (ICLR), 2026.
- Jiong Zhu*, Gaotang Li*, **Yao-An Yang**, Jing Zhu, Xuehao Cui, Danai Koutra. 2024. On the impact of feature heterophily on link prediction with graph neural networks. Advances in Neural Information Processing Systems (NeurIPS), 2024. (*equal contributions)

Lab Affiliation

- **Situated Language and Embodied Dialogue (SLED) lab** **Michigan, USA**
 - Student Researcher* *May 2024 – Present*
 - Advisor: Prof. Joyce Chai
 - Collaborators: Ziqiao Ma, Prof. Jonathan Brennan, Prof. Freda Shi
 - Research topics: Vision Language Model steering and grounding, LLM pattern learning and habituation
- **Graph Exploration and Mining at Scale (GEMS) lab** **Michigan, USA**
 - Student Researcher* *Mar 2023 – Present*
 - Advisor: Prof. Danai Koutra
 - Collaborators: Donald Loveland, Gaotang Li, Dr. Jiong Zhu
 - Research topics: Graph Neural Networks, Graph Learning and link prediction under Heterophily, Text Attributed Graph Learning through GNN and LLM fusion
- **Lorenzon Lab (Michigan Xenon Group)** **Michigan, USA**
 - Student Researcher* *Oct 2023 – Nov 2024*
 - Advisor: Prof. Wolfgang Lorenzon
 - Collaborators: Dr. Gregory Rischbieter
 - Research topics: Hardware development for cryogenic Xenon particle detector, data analytics with Python, data acquisition with C++

Research Projects

- **Vision Language Model (VLM) Grounding through Steering** **Michigan, USA**
 - Project Lead* *Jun 2025 – Present*
 - Designed algorithm to improve VLM grounding and attention to detail through lightweight steering head
 - Developed code for dataloading, VLM model and steering head, VLM training and finetuning, and evaluation
 - In collaboration with Ziqiao Ma and Prof. Joyce Chai
- **Probing Large LLM Working Memory through Patterned Stimuli** **Michigan, USA**
 - Co-Project Lead* *Jan 2025 – Present*
 - Comparing language models and humans working memory on patterned stimuli
 - Developed code evaluating large language model on repeated and patterned textual input
 - Designed pipeline to test human subject reaction time on repeated and patterned text
 - In collaboration with Ziqiao Ma, Junyuan Zhao, Prof. Jonathan Brennan, Prof. Freda Shi and Prof. Joyce Chai

- **Designing Effective GNN-LLM Fusion** **Michigan, USA**
Jun 2025 – Sep 2025
 - *Co-Author*
 - Benchmarking and improving efficiency and effectiveness of graph learning pipelines that use LLM
 - Implemented LLM, GNN, and fusion models for comparison on datasets ranging from 5K to 6M edges
 - Created pipelines to finetune language models for embedding text for graph learning tasks
 - Work under review at the International Conference on Learning Representations (ICLR) 2026
- **Evaluating Effective Designs for Heterophilic Link Prediction with GNN** **Michigan, USA**
Mar 2023 – May 2024
 - *Co-Author*
 - Implemented dataloading pipelines and GNN models such as GCN, SAGE and BUDDY
 - Ran ablation studies, benchmarked models, and analyzed experiment results presented in the paper
 - Identified key design choices for GNNs to effectively perform link prediction under heterophily
 - Work published in the Advances in Neural Information Processing Systems (NeurIPS) 2024

Work Experience

- **ViaInno Corporation** **Taipei, Taiwan**
Jul 2023 – Jul 2024
 - *Software Developer*
 - Worked in-person full-time over summer and remote part-time during the school year
 - Developed mobile app features for large apartment buildings management using Flutter
 - Tested web apps written in Vue.js and mobile apps written in Flutter
 - Redesigned company website using Tailwind CSS and Vue.js for product marketing

Extracurriculars

- **Michigan Hackathon (MHacks)** **Michigan, USA**
Nov 2023
 - *First Place Winner*
 - Developed Flutter mobile app for users to sign in with Google OAuth and upload videos to Firebase database
 - Generated 3D models of the uploaded videos using Neural Radiance Fields (NeRF)
 - Created a 3D environment for users to scale and move generated 3D models for interior design
- **Michigan Robotic Submarine Team** **Michigan, USA**
Sep 2022 – Oct 2023
 - *Software Team Engineer*
 - Implemented and tested state estimation using Extended Kalman Filters
 - Worked on real-time underwater object detection using OpenCV and YOLO
 - Designed state machine diagrams for task planning and motor control testing
- **Michigan Autonomous Aerial Vehicle Team** **Michigan, USA**
Sep 2022 – Jul 2023
 - *Embedded System Team Engineer*
 - Designed and assembled printed circuit boards using Autodesk Eagle
 - Developed safety features and emergency stop for autonomous drone
 - Implemented embedded code for STM32 microcontroller

Awards and Honors

- **University Honors (All 6 semesters), University of Michigan** **Dec 2022 - May 2025**
- **James B. Angell Scholar (2 time recipient), University of Michigan** **Mar 2024 & Mar 2025**
- **NeurIPS Scholar Award, Conference on Neural Information Processing Systems** **Dec 2024**
- **William J. Branstrom Freshman Prize (top 5% freshman), University of Michigan** **Mar 2023**